EV Technology Course — Module 6: High-Voltage and Low-Voltage Electrical System

Oct 14, 2026 · @Ashishpreet

An EV carries two electrical networks: a high-voltage (HV) system that moves tens of kilowatts at 300–800 V, and a 12 V (increasingly 48 V) low-voltage system that runs everything else. The HV system is built to be safe even when something fails: it is isolated from the car body, interlocked, fused and constantly monitored. This module covers its layout, cables, connectors and safety concepts, and the LV network that keeps the car alive.

1. Learning objectives

- Lay out the HV distribution chain from pack to every HV load.
- Explain isolation, insulation monitoring, the HV interlock loop (HVIL) and equipotential bonding.
- Size an HV cable for current, voltage drop and heat; compare copper and aluminium.
- Describe HV connectors, busbars, charging inlets and their tests.
- Explain the 12 V network, the DC-DC converter and the move to 48 V and zonal power distribution.

2. Fundamentals

Why HV? Power = voltage × current. Moving 100 kW at 12 V would need 8,300 A; at 400 V it needs 250 A; at 800 V, 125 A. Higher voltage means thinner cables and lower losses.

Why keep 12 V? Lamps, ECUs, sensors, airbags and infotainment are built for 12 V; it is safe to touch; and it must work even when the HV system is switched off (for unlocking, waking up and safety functions).

Safety by design. Automotive rules treat anything above 60 V DC as high voltage. EV HV systems are **floating** (not connected to the car body). One insulation fault therefore causes no shock; the system detects it and warns before a second fault can make a dangerous path.

3. Detailed technical explanation

HV components

Component	Function
Main fuse / pyrofuse	Breaks short-circuit currents; pyrofuse fires on crash signal
Contactors + pre-charge	Connect and disconnect the pack safely (battery course Module 6)
HV junction box / PDU	Distributes power to loads through busbars and branch fuses
HV cables	Orange, shielded, double-insulated; copper or aluminium conductors
HV connectors	Touch-safe, sealed (IP67/IP6K9K), keyed, with HVIL pins and shield contacts
Charging inlet	AC/DC contacts, temperature sensors, locking actuator, communication pins
Insulation monitoring device (IMD)	Measures resistance between HV and chassis continuously
HVIL	Low-voltage loop through every HV connector and cover; opening it opens the contactors
Manual service disconnect	Physical break for technicians
Equipotential bonding	Bonds all HV component housings to the chassis so touchable metal stays at the same potential

HV loads: traction inverter(s), onboard charger, DC-DC converter, electric AC compressor, PTC heaters, sometimes electric power steering on large vehicles.

EMC. Fast-switching inverters create electrical noise. Shielded cables with 360° shield termination, filters and careful routing stop interference with radios, sensors and ADAS.

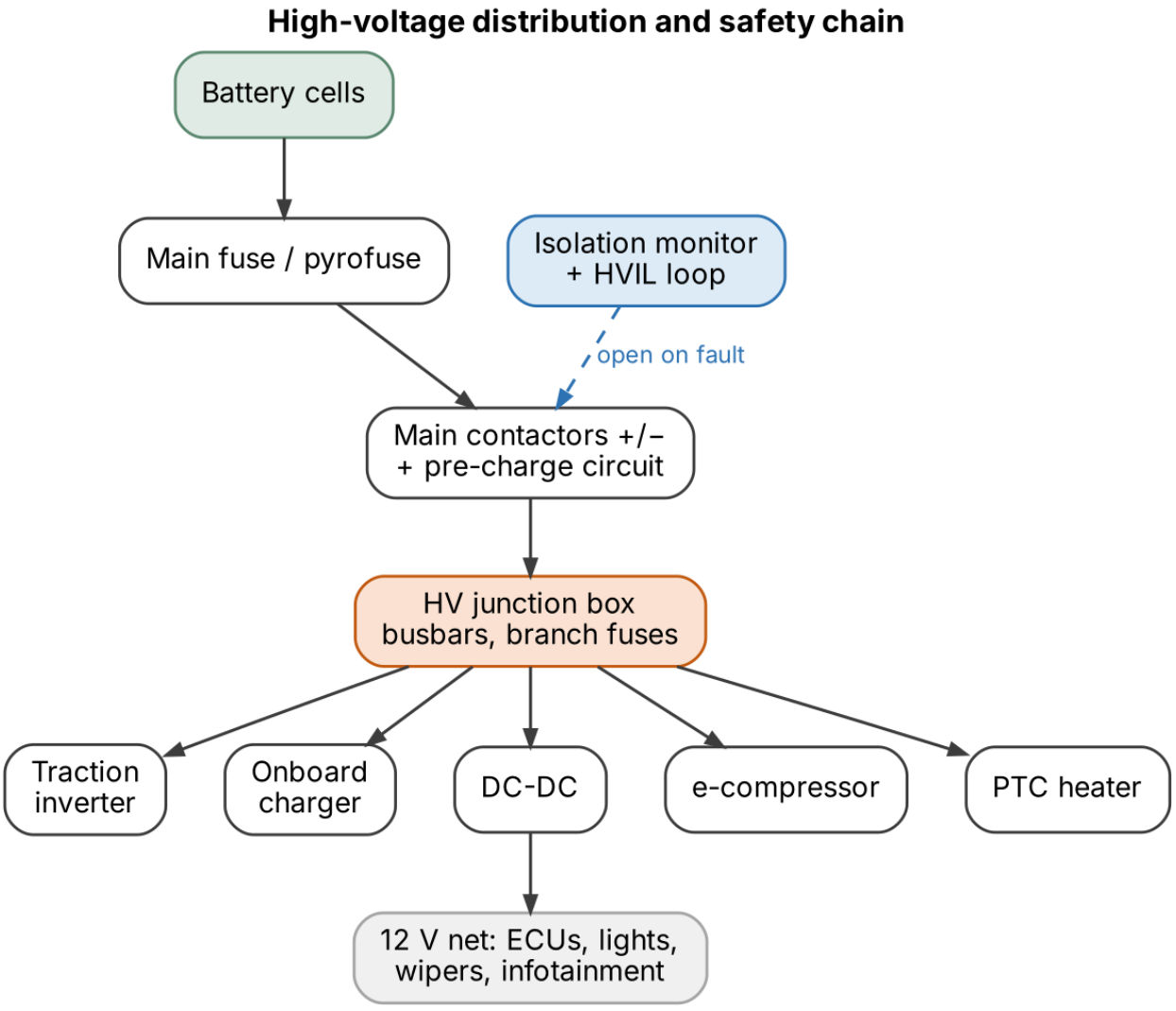
Low-voltage system

Element	Notes
12 V battery	Lead-acid or increasingly LFP; powers wake-up and safety functions
DC-DC converter	Replaces the alternator; keeps the 12 V battery charged when HV is on

Element	Notes
Power distribution	Fuse boxes moving to electronic fuses (eFuses) and smart power distribution
48 V	Lower current for high loads; Tesla Cybertruck uses a 48 V LV architecture
Zonal distribution	Zone controllers distribute power and data locally, shortening the harness (Module 10)

Harness facts. A modern car harness can contain several kilometres of wire and weigh tens of kilograms. Zonal architectures and higher LV voltage aim to cut this.

4. Diagram: HV distribution and safety chain



5. Step-by-step: how the system handles an insulation fault

1. Rain water reaches a damaged HV connector; insulation resistance between HV+ and chassis drops.
2. The IMD, which constantly measures this resistance, sees it fall below the warning threshold.
3. The BMS/VCU logs the fault and warns the driver; the car stays drivable because a single fault is not a shock hazard in a floating system.
4. If resistance falls below the critical threshold, the vehicle prevents the next HV start (or, depending on strategy, reduces power and asks the driver to stop).
5. A technician isolates the HV system (service disconnect, lock-out), verifies zero voltage and repairs the connector.

6. Engineering equations

Conductor resistance, voltage drop and loss:

$$R = \frac{\rho L}{A} \qquad \Delta V = I R \qquad P = I^2 R$$

Minimum isolation resistance (UNECE R100 / AIS-038):

$$R_{iso} \geq 100\ \Omega/\text{V} \times V_{max}\ (\text{DC circuits}) \qquad R_{iso} \geq 500\ \Omega/\text{V} \times V_{max}\ (\text{with AC circuits not separately protected})$$

Power and current at a given voltage:

$$I = \frac{P}{V}$$

ρ = resistivity (copper 1.72×10^{-8} Ω·m; aluminium 2.82×10^{-8} Ω·m), L = length (m), A = cross-section (m²).

7. Worked examples

Example 1: copper HV cable. 35 mm², 3 m, 300 A: R = $1.72 \times 10^{-8} \times 3 \div 35 \times 10^{-6}$ = **1.47 mΩ**; drop **0.44 V**; heat **133 W**.

Example 2: aluminium alternative. 50 mm² aluminium, same length: R = **1.69 mΩ**, heat **152 W**, but conductor mass is about **405 g** against **941 g** for the copper cable. Aluminium needs a bigger section but saves weight and cost; terminations need special processes (oxide, creep).

Example 3: isolation threshold. For a 400 V DC system: 100 × 400 = **40 kΩ** minimum; if AC circuits are galvanically connected without extra protection: 500 × 400 = **200 kΩ**.

Example 4: why 48 V. A 1.5 kW load at 14 V draws **107 A**; at 48 V, **31 A**. Cable losses fall by the square of current ratio: about **12×** lower for the same cable.

8. Real-world examples

- All Indian EV cars use orange HV cabling, HVIL and insulation monitoring as required by AIS-038 Rev 2.
- **Tesla Cybertruck:** 48 V low-voltage architecture and Ethernet-based backbone.
- **800 V cars (Hyundai E-GMP, Porsche):** need higher-rated insulation, connectors and creepage distances; some include a DC boost or switching to charge on 400 V chargers.
- **Two-wheelers:** 48–72 V systems; many stay near or below 60 V DC to simplify safety requirements.

9. Manufacturing applications

- **Harness manufacturing:** cutting, stripping, crimping or ultrasonic welding of terminals; shield termination; 100% electrical continuity and hi-pot tests.
- **HV assembly in the plant:** torque-controlled HV bolts with recorded values, connector seating and latch checks, HVIL continuity, insulation resistance test of the full vehicle, equipotential bonding resistance check (typically below 0.1 Ω).
- **Qualified workforce:** operators handling live HV need training and authorisation; most plant steps are done with HV not yet live.
- **Costing:** copper price drives cable cost; aluminium cables, busbars and shorter zonal harnesses reduce cost and mass.

10. Common mistakes and misconceptions

- **"HV is connected to the body like 12 V."** HV is floating; the body is not part of the HV circuit.
- **"Switching the car off makes HV safe."** The pack itself stays charged; only the contactors open.
- **"Thicker cable is always safer."** It adds mass and cost; size for current, temperature rise and short-circuit duty.
- **"12 V failures are minor."** A flat 12 V battery prevents the HV system from starting.

11. Practical case study: intermittent isolation warnings in monsoon

Problem. A batch of cars shows isolation warnings after heavy rain, which clear after drying.

Root cause. A connector seal was damaged during assembly because of insufficient lead-in on the mating part, combined with routing that let water pool around the connector.

Corrective action. Redesign the routing with a drip loop, add a seal inspection step and connector push-pull test, update the PFMEA, and run water-ingress validation (IPX7/IPX9K) on the full vehicle, not just components.

Lesson. HV safety depends on assembly quality and routing, not only component ratings.

12. Revision notes

- HV = floating, isolated, interlocked, fused, monitored.
- IMD checks isolation; HVIL opens contactors when a connector opens.
- Cable sizing: $R = \rho L \div A$; heat = I^2R.
- Isolation minimum 100 Ω/V (DC) or 500 Ω/V (unprotected AC).
- 12 V stays; 48 V and zonal power are coming.

13. Glossary

Term	Meaning
Floating (IT) system	HV not referenced to chassis ground
IMD	Insulation monitoring device
HVIL	High-voltage interlock loop
PDU	Power distribution unit / junction box
Equipotential bonding	Electrical bonding of metal housings to chassis
eFuse	Electronic fuse with programmable trip behaviour
Creepage / clearance	Insulation distance along a surface / through air

14. Knowledge check

Conceptual

1. Why is a single insulation fault not a shock hazard in a floating HV system?
2. What is the role of the HVIL?
3. Why does an EV still need a 12 V battery?
4. What are the pros and cons of aluminium HV cables?
5. Why does EMC matter more in EVs?

Numerical

1. 25 mm² copper cable, 2 m, 200 A: resistance and heat?
2. Minimum isolation resistance for an 800 V DC system?
3. A 1 kW LV load: current at 13.5 V and at 48 V?

Application

1. Draft the HV-related end-of-line tests for an EV assembly line.
2. An OEM wants to cut harness mass by 20%. Suggest options.

15. Answer key

Conceptual

1. There is no return path through the body; a person touching one faulty conductor and the body does not complete a circuit. A second fault on the other pole would.
2. It detects any open HV connector or removed cover and opens the contactors so no one is exposed to live terminals.
3. ECUs, lights, locks and safety systems run on 12 V and must work to wake the car and close the contactors; it also powers the car when HV is off.
4. Pros: about half the mass and lower cost. Cons: larger cross-section, harder termination (oxide, creep, galvanic corrosion), lower flex fatigue life.
5. Fast-switching inverters at hundreds of volts create strong electromagnetic noise that can disturb radios, sensors and ADAS.

Numerical

1. $R = 1.72 \times 10^{-8} \times 2 \div 25 \times 10^{-6}$ = **1.38 mΩ**; heat = $200^2 \times 0.00138$ = **55 W**.
2. 100 × 800 = **80 kΩ**.
3. **74 A** at 13.5 V; **21 A** at 48 V.

Application

1. Equipotential bonding resistance, HV insulation resistance (vehicle level), HVIL continuity, contactor and pre-charge function, charging inlet communication and lock test, AC and DC charging function test, HV connector torque and latch verification records, DC-DC output check.

2. Zonal architecture to shorten runs; aluminium for large cross-sections; 48 V for high-current LV loads; eFuses instead of oversized wires; integrated power electronics to remove HV cable runs; optimise routing with harness design software.

16. Further learning

- UNECE Regulation No. 100; ISO 6469-3 (electrical safety); AIS-038 Rev 2.
- LV 214/LV 215 connector test standards (German OEMs).
- Bosch Automotive Electrics and Automotive Electronics handbook.
- SAE J1673 (HV automotive wiring).

Next: Module 7, vehicle thermal management.